

Template Screening Report (v5.7.1 - Multi-Source AI Evaluator)

Journal Template: Jurnal Teknologi

Submission No: 26799

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Article Title: VERIFICATION OF SHUTDOWN MARGIN AND EXCESS REACTIVITY VALUES ON GAS-RSG EQUILIBRIUM CORE AFTER 35 YEARS OF OPERATIONS

Source File: JT-tukiran-utm2026.docx

Screened PDF: JT-tukiran-utm2026.pdf

Screening Date: 2026-05-21 09:40:04

Detection Source Debug Report

Item	DOC Detection	PDF Detection	AI Detection	Final Source
Article Title	NO	YES	YES	PDF + AI
Authors	NO	YES	YES	PDF + AI
Affiliations	NO	YES	YES	PDF + AI
Corresponding Author	YES	YES	SKIPPED	DOC + PDF
Graphical Abstract	YES	YES	SKIPPED	DOC + PDF
English Abstract	YES	YES	SKIPPED	DOC + PDF
Keywords	YES	YES	SKIPPED	DOC + PDF
Malay Abstract (Abstrak)	NO	YES	YES	PDF + AI
Kata Kunci	NO	NO	NO	None
Introduction	YES	NO	YES	DOC + AI
Methodology	YES	NO	YES	DOC + AI
Results and Discussion	YES	YES	SKIPPED	DOC + PDF
Conclusion	YES	NO	YES	DOC + AI
Acknowledgement	YES	YES	SKIPPED	DOC + PDF
References	YES	YES	SKIPPED	DOC + PDF

Summary: DOC detected 10 items | PDF detected 11 items | AI confirmed 7 items

The system uses DOC + PDF + AI voting logic. AI serves as a third checker when DOC and PDF conflict. AI is skipped when DOC and PDF already agree.

Executive Summary

Overall Decision: **NEEDS MANUAL REVIEW**

Main Reason: The manuscript generally follows the template but contains inconsistencies that require manual review.

Critical Failed Items: None

Items Needing Review:

- Article Title

- Authors
- Affiliations
- Malay Abstract (Abstrak)
- Introduction

Suggested Action: Editor should review the highlighted issues before proceeding.

Overall Result

Metric	Score / Value
Final Status	NEEDS MANUAL REVIEW
Final Score	76/100
Structure Score	47%
Layout Score	89%
Formatting Score	100%
Detection Method	DOCX-First + Multi-Source AI (v5.7.1)
Decision	MANUAL REVIEW

How Items Are Evaluated

DOC + PDF + AI SKIPPED	PASS	Both deterministic detectors agree
DOC + PDF + AI CONFIRMED	PASS	All three confirm
DOC + AI	NEEDS REVIEW	DOC+AI confirm, but PDF layout conflicts
PDF + AI	NEEDS REVIEW	PDF+AI confirm, but DOC source extraction fails
DOC only	NEEDS REVIEW	DOC detects, but PDF+AI don't confirm
PDF only	NEEDS REVIEW	PDF detects, but DOC+AI don't confirm
AI only	NEEDS REVIEW	Only AI confirms (deterministic detectors failed)
None	FAILED	All three fail to detect required item

Status Explanation

PASS — Item follows the template structure, position, and formatting.

WARNING/NEEDS REVIEW — Item exists but evidence is inconsistent or layout differs from template.

FAILED — Required item truly missing after DOC/PDF/AI checks.

Item Screening Details

Item	DOC	PDF	AI	Final Source	Status	Reason
Article Title	NO	YES	YES	PDF + AI	WARNING	Detected by PDF and AI, but DOC source text did not co
Authors	NO	YES	YES	PDF + AI	WARNING	Detected by PDF and AI, but DOC source text did not
Affiliations	NO	YES	YES	PDF + AI	WARNING	Cons detected by PDF and AI, but DOC source text did
Corresponding Author	YES	YES	SKIPPED	DOC + PDF	PASS	Corresponding_Author confirmed by both DOC and PDF detection

Graphical Abstract	YES	YES	SKIPPED	DOC + PDF	PASS	Graphical_Abtract confirmed by both DOC and PDF detection.
English Abstract	YES	YES	SKIPPED	DOC + PDF	PASS	Abstract confirmed by both DOC and PDF detection.
Keywords	YES	YES	SKIPPED	DOC + PDF	PASS	Keywords confirmed by both DOC and PDF detection.
Malay Abstract (Abstrak)	NO	YES	YES	PDF + AI	WARNING	Abstract detected by PDF and AI, but DOC source text did not
Kata Kunci	NO	NO	NO	None	FAILED	Kunci not found by DOC, PDF, or AI. [AI: weak_evidence,
Introduction	YES	NO	YES	DOC + AI	WARNING	Introduction confirmed by DOC and AI, but PDF/layout detecti
Methodology	YES	NO	YES	DOC + AI	WARNING	Methodology confirmed by DOC and AI, but PDF/layout detectio
Results and Discussion	YES	YES	SKIPPED	DOC + PDF	PASS	Results_Discussion confirmed by both DOC and PDF detection.
Conclusion	YES	NO	YES	DOC + AI	WARNING	Conclusion confirmed by DOC and AI, but PDF/layout detection
Acknowledgement	YES	YES	SKIPPED	DOC + PDF	PASS	Acknowledgement confirmed by both DOC and PDF detection.
References	YES	YES	SKIPPED	DOC + PDF	PASS	References confirmed by both DOC and PDF detection.

Detailed Findings

Article Title [title]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Article Title' section must exist.
Found: Matched heading: "VERIFICATION SHUTDOWN MARGIN AND EXCESS REACTIVITY VALUES ON GAS-RSG EQUILIBRIUM CORE AFTER " | Detection method: score_rank
Confidence: 1.00
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 1.00
AI Evidence: PDF snippet shows the full title and author information clearly, while DOC snippet does not contain the title section at all.
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Rejected false positives: Jurnal Teknologi, Full Paper, Article history
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, top area
Found: Page 1, left column, upper area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.10).

C. Formatting Evaluation

Expected: Heading should be in uppercase style.
Expected: Heading should use bold font.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Authors [authors]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Authors' section must exist.
Found: Matched heading: "VERIFICATION SHUTDOWN MARGIN AND EXCESS REACTIVITY VALUES ON GAS-RSG EQUILIBRIUM CORE AFTER " | Detection method: score_rank
Confidence: 0.70
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 0.85
AI Evidence: PDF snippet shows a full authors section with multiple authors and affiliations; DOC snippet does not contain any authors section, only other manuscri
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Rejected false positives: Article history, Received, Accepted
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, left column, upper area
Position Check: PASS
Reason: Position matches expected template region.

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Affiliations [affiliations]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Affiliations' section must exist.

Found: Matched heading: "Nuclear Energy, National Research and Innovation Agency (BRIN), KST BJ Habiebi, Serpong Gd. 80, Tan" | Detection method: score_rank

Confidence: 0.70

Detection Method: score_ranking

DOC Detection: NO

PDF Detection: YES

AI Detection: YES

AI Confidence: 0.85

AI Evidence: PDF snippet shows full Affiliations section with author names and institutional addresses. DOC snippet does not contain the Affiliations section, only

Evidence Sources: PDF + AI

Final Source Used: PDF + AI

Rejected false positives: Accepted

Status: WARNING

Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area

Found: Page 1, left column, upper area

Position Check: PASS

Reason: Position matches expected template region.

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Corresponding Author [corresponding_author]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Corresponding Author' section must exist.

Found: Matched heading: "Corresponding author" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: YES

AI Detection: SKIPPED

Evidence Sources: DOC + PDF

Final Source Used: DOC + PDF

Status: PASS

Reason: The required component 'Corresponding Author' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area

Found: Page 1, right-center area, upper area

Position Check: WARNING

Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: **PASS**

Graphical Abstract [graphical_abstract]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Graphical Abstract' component must exist.

Found: Matched heading: "Graphical abstract" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: YES

AI Detection: SKIPPED

Evidence Sources: DOC + PDF

Final Source Used: DOC + PDF

Status: **PASS**

Reason: The required component 'Graphical Abstract' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, upper area

Found: Page 1, left-center area, upper-middle area

Position Check: **WARNING**

Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Image should be present.

Formatting Check: **PASS**

Reason: Formatting matches template expectations.

Overall Item Status: **PASS**

English Abstract [abstract]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'English Abstract' section must exist.

Found: Matched heading: "Abstract" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: YES

AI Detection: SKIPPED

Evidence Sources: DOC + PDF

Final Source Used: DOC + PDF

Status: **PASS**

Reason: The required component 'English Abstract' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area

Found: Page 1, left-center area, upper-middle area

Position Check: **WARNING**

Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: **PASS**

Reason: Formatting matches template expectations.

Overall Item Status: **PASS**

Keywords [keywords]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Keywords' section must exist.

Found: Matched heading: "Keywords" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: YES

AI Detection: SKIPPED

Evidence Sources: DOC + PDF

Final Source Used: DOC + PDF

Status: **PASS**

Reason: The required component 'Keywords' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, lower-middle area

Found: Page 1, left-center area, lower-middle area

Position Check: **WARNING**

Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: **PASS**

Reason: Formatting matches template expectations.

Overall Item Status: **PASS**

Malay Abstract (Abstrak) [abstrak]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Malay Abstract (Abstrak)' section must exist.

Found: Nearby content: "78:1 (2016) 1–5 | www.jurnalteknologi.utm.my | eISSN 2180–3722 | Jurnal Teknologi Full Paper
VERIFICATION SHUTDOWN MARGIN AND EXC

Confidence: 0.00

Detection Method: semantic_fallback

DOC Detection: NO

PDF Detection: YES

AI Detection: YES

AI Confidence: 0.70

AI Evidence: PDF snippet shows manuscript metadata and partial content consistent with presence of Abstrak; DOC snippet is missing this content entirely.

Evidence Sources: PDF + AI

Final Source Used: PDF + AI

Status: **WARNING**

Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, lower area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: **PASS**

Reason: Formatting matches template expectations.

Overall Item Status: **WARNING**

Kata Kunci [kata_kunci]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Kata Kunci' section must exist.

Found: Detection method: not_found

Confidence: 0.00

Detection Method: not_found

DOC Detection: NO

PDF Detection: NO
AI Detection: NO
AI Evidence: DOC snippet includes title, authors, affiliations, and article history but no keywords. PDF snippet similarly includes title, authors, affiliations, a
Evidence Sources: None
Final Source Used: None
Status: **FAILED**
Reason: Required component 'Kata Kunci' was not found.
B. Layout / Coordinate Evaluation
Expected: Page 1, left column, lower area
Found: Not detected
C. Formatting Evaluation
Expected: Standard paragraph formatting.
Formatting Check: **PASS**
Reason: Formatting matches template expectations.
Overall Item Status: **FAILED**

Introduction [introduction]

A. Structure Evaluation (DOC + PDF + AI)
Expected: The 'Introduction' section must exist.
Found: Matched heading: "1.0. INTRODUCTION" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: NO
AI Detection: YES
AI Confidence: 0.85
AI Evidence: DOC snippet provides detailed introduction content about the reactor, regulatory context, and study objectives. PDF snippet lacks Introduction content
Evidence Sources: DOC + AI
Final Source Used: DOC + AI
Rejected false positives: 1.0. INTRODUCTION
Status: **WARNING**
Reason: Component found but evidence is inconsistent.
B. Layout / Coordinate Evaluation
Expected: Page 1, left column, top area
Found: Not detected
C. Formatting Evaluation
Expected: Standard paragraph formatting.
Formatting Check: **PASS**
Reason: Formatting matches template expectations.
Overall Item Status: **WARNING**

Methodology [methodology]

A. Structure Evaluation (DOC + PDF + AI)
Expected: The 'Methodology' section must exist.
Found: Matched heading: "3.0. METHODOLOGY" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: NO
AI Detection: YES
AI Confidence: 0.85
AI Evidence: DOC snippet includes detailed methodology content about core calculations and modeling using the WIMSD-5B program. PDF snippet only contains header an

Evidence Sources: DOC + AI
Final Source Used: DOC + AI
Rejected false positives: 3.0. METHODOLOGY
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Results and Discussion [results_discussion]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Results and Discussion' section must exist.
Found: Matched heading: "4.0. RESULTS AND DISCUSSIONS" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS

Reason: The required component 'Results and Discussion' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Page 1, center area, lower area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: PASS

Conclusion [conclusion]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Conclusion' section must exist.
Found: Matched heading: "4.0. CONCLUSION" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: NO
AI Detection: YES
AI Confidence: 0.85
AI Evidence: DOC snippet includes a detailed Conclusion discussing verification calculations, safety parameters, and operational limits. PDF snippet lacks any Conc
Evidence Sources: DOC + AI
Final Source Used: DOC + AI
Rejected false positives: 4.0. CONCLUSION
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: **PASS**

Reason: Formatting matches template expectations.

Overall Item Status: **WARNING**

Acknowledgement [acknowledgement]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Acknowledgement' component is recommended.

Found: Matched heading: "Acknowledgment" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: YES

AI Detection: SKIPPED

Evidence Sources: DOC + PDF

Final Source Used: DOC + PDF

Status: **PASS**

Reason: The required component 'Acknowledgement' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area

Found: Page 1, left-center area, upper-middle area

Position Check: **WARNING**

Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: **PASS**

Reason: Formatting matches template expectations.

Overall Item Status: **PASS**

References [references]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'References' section must exist.

Found: Matched heading: "References" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: YES

AI Detection: SKIPPED

Evidence Sources: DOC + PDF

Final Source Used: DOC + PDF

Status: **PASS**

Reason: The required component 'References' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area

Found: Page 1, left column, lower-middle area

Position Check: **WARNING**

Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: **PASS**

Reason: Formatting matches template expectations.

Overall Item Status: PASS

Editor Conclusion

The manuscript generally follows the template but has items that require manual review. These items had inconsistent evidence between DOC detection, PDF layout analysis, or AI verification. Please review the highlighted items before making a final decision.

Structure Score: 47% (Weight: 40%)

Layout Score: 89% (Weight: 30%)

Formatting Score: 100% (Weight: 30%)

Final Score: 76/100

Items Needing Review:

- Article Title — **WARNING**
- Authors — **WARNING**
- Affiliations — **WARNING**
- Malay Abstract (Abstrak) — **WARNING**
- Introduction — **WARNING**

This report was automatically generated by Turnitin Assistant v5.7.1 Template Screening Engine (DOCX-First + Multi-Source AI Evaluator).